

MATH102 SMP Week 4

2020F Mid Problem 3

Is the function $f(x, y) = 1 + |x + y| + xy^2$ differentiable at the point $(2, -1)$? Justify your answer.

Textbook 2.49

Let $0 < b < a$ and consider the set of points in the plane whose polar coordinates r, θ satisfy

$$r = \frac{1}{a + b \cos \theta}$$

(a) Show that the x, y coordinates of the points satisfy

$$1 = a\sqrt{x^2 + y^2} + bx$$

(b) Show that the equation $1 = a\sqrt{x^2 + y^2} + bx$ is the equation of an ellipse by completing a square and getting the equation into the form

$$1 = \frac{(x - \alpha)^2}{c^2} + \frac{(y - \beta)^2}{d^2}$$

Textbook 3.2

Let $f(x, y) = x^2 + 3y$. Find the linear approximation of $f(x, y)$ near $(2, 4)$, and use it to estimate $f(2.01, 4.03)$.

Textbook 3.29

Suppose $u(x, y)$ is a C^2 function from $\mathbb{R}^2$ to $\mathbb{R}$. After a change to polar coordinates $u(x, y) = u(r \cos \theta, r \sin \theta)$, use the Chain Rule to show that

$$u_{xx} + u_{yy} = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}$$

Textbook 3.31 (d)

Suppose p , q , and r are C^2 functions such that

$$p_x = q_y, \quad p_y = -q_x, \quad r_{xx} + r_{yy} = 0$$

Define $w(x, y) = r(p(x, y), q(x, y))$. Show that $w_{xx} + w_{yy} = 0$.

Textbook 3.35

Let $\mathbf{F}(x, y) = (e^x \cos y, e^x \sin y)$

- (a) Find the derivative matrix $D\mathbf{F}$.
- (b) Show that $D\mathbf{F}(\mathbf{A})^{-1}$ exists at every point $\mathbf{A} = (a, b)$ so $\mathbf{F}$ is locally invertible at $\mathbf{A}$.
- (c) Show that $\mathbf{F}$ is not invertible globally; that is, find different points of the domain that are mapped to the same value in the range.

Textbook 3.37

Let $\mathbf{F}(x, y) = (x^4 + 2xy + 1, y)$, and consider the equations

$$\begin{aligned}x^4 + 2xy + 1 &= y \\ y &= v\end{aligned}$$

that is $\mathbf{F}(x, y) = (u, v)$.

- (a) Show that for $(u, v) = (0, 1)$, one solution is $(x, y) = (-1, 1)$.
- (b) Find the derivative matrix $D\mathbf{F}(x, y)$.
- (c) Show that $D\mathbf{F}(-1, 1)$ is invertible, and hence $\mathbf{F}^{-1}$ exists on a small enough disk centered at $(0, 1)$.
- (d) Use the linear approximation to estimate $\mathbf{F}^{-1}(0.2, 1.01)$.

Textbook 3.40

Let $f(x, y, z) = z^4 + 2yz + x$, and consider the level set

$$z^4 + 2yz + x = 0$$

- (a) Show that $(0, 2, 0)$ is on the level set.
- (b) Find the partial derivatives of f .
- (c) Show that there is a function $z = g(x, y)$ defined near $(0, 2)$ so that

$$f(x, y, g(x, y)) = 0$$

- (d) Find the partial derivatives $g_x(x, y)$ and $g_y(x, y)$, and use them to find the second derivatives $g_{xx}(0, 2)$, $g_{xy}(0, 2)$, $g_{yy}(0, 2)$.

Textbook 3.41

Let

$$f_1(x, y_1, y_2) = 3y_1 + y_2^2 + 4x = 0$$

$$f_2(x, y_1, y_2) = 4y_1^3 + y_2 + x = 0$$

- (a) Verify that $(x, y_1, y_2) = (-4, 0, 4)$ is a solution.
- (b) Show that there is a function $\mathbf{G} = (g_1, g_2)$ such that $g_1(x) = y_1$, $g_2(x) = y_2$ for all solutions near $(-4, 0, 4)$.
- (c) Show that at $x = -4$,

$$\frac{dg_1}{dx} = \frac{4}{3}, \quad \frac{dg_2}{dx} = -1$$

Textbook 3.42

Let $\mathbf{F}(u, v, w) = (u + v + w, uv + vw + wu)$, and consider the system of equations $\mathbf{F}(u, v, w) = (2, -4)$:

$$u + v + w = 2$$

$$uv + vw + wu = -4$$

- (a) Verify that $(2, 2, -2)$ is a solution.
- (b) Writing $\mathbf{F}(u, v, w) = (f_1(u, v, w), f_2(u, v, w))$, find all the first partial derivatives of f_1 and f_2 .
- (c) Verify that $D\mathbf{F}(2, 2, -2) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 4 \end{bmatrix}$.
- (d) Can the equations be solved for v and w in terms of u near $(2, 2, -2)$? That means: does the Implicit Function Theorem guarantee existence of a function $\mathbf{G} = (g_1, g_2)$ defined on some interval around 2 so that $\mathbf{G}(2) = (2, -2)$ and $\mathbf{F}(u, g_1(u), g_2(u)) = (2, -4)$?
- (e) Can the equations be solved for u and v in terms of w near $(2, 2, -2)$?